

Supplemental Material for “Nanoscale Critical Phenomena in a Complex Fluid Studied by X-ray Photon Correlation Spectroscopy”

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This supplemental material gives additional detail on Sample Preparation, Characterization Methods, Fitting Methodology, Analysis of Time-Averaged X-ray Data, and XPCS Data Analysis.

SAMPLE PREPARATION

The purified extractant DMBTDMMA (N,N'-dimethyl N,N'-dibutyl tetra-*n*-decyl-malonamide, see Fig.S.1) was obtained from Technocomm Ltd and used as received (99 % by HPLC). Cerium(III) nitrate hexahydrate and *n*-dodecane were obtained from Sigma-Aldrich. Optima Grade nitric acid was obtained from Fisher Scientific. The initial aqueous phase consisted of a 0.01 M Ce(NO₃)₃ solution in 3 M HNO₃ (prepared using 18 MΩ cm water with less than 1 ppb total organic content). The initial organic phase was a 0.4 M DMBTDMMA solution in *n*-dodecane. Equal volumes of the initial aqueous and organic phases in a screw-cap glass test tube were shaken at 23 ± 0.5 °C for 20 minutes using an IKA MS3 digital mixer (1500 min⁻¹), followed by 10 minutes centrifugation (Drucker Co. model 614B). After mixing and settling, the equilibrated organic phase was physically separated from the aqueous phase and a small aliquot (approx. 200 μL) was injected into a 4-mm-diameter quartz capillary tube (Charles Supper, QZ-40) for variable-temperature XPCS measurements.

Liquid-liquid phase separation in the organic phase occurs at lower *T* or higher solute loadings [1–3]. We chose the initial aqueous phase concentration (0.01 M Ce(NO₃)₃·3 M HNO₃) to be just below the concentration

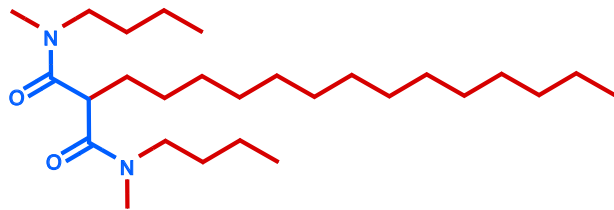


FIG. S.1. The structure of the neutral molecular amphiphile DMBTDMMA extractant showing the hydrophilic head group (blue) and oleophilic tail groups (red).

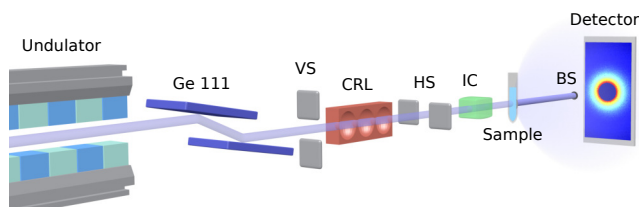


FIG. S.2. Schematic view of XPCS setup at 8ID-I. VS - vertical slits, CRL - compound refractive lenses, HS - horizontal slits, IC - ion chamber, BS - beam stop.

that leads to organic phase separation at room temperature [4, 5]. We chose the initial organic phase concentration (0.4 M DMBTDMMA-*n*-dodecane) to be at or near the critical concentration [6–8]. The critical organic concentration was determined by varying the DMBTDMMA concentration between 0.1 and 0.6 M in 0.1 M steps; the 0.4 M sample showed the highest temperature of phase splitting.

CHARACTERIZATION METHODS

X-ray photon correlation spectroscopy

The X-ray photon correlation spectroscopy (XPCS) experiment was carried out at beamline 8ID of the Advanced Photon Source (APS). A sketch of the XPCS experimental setup is shown in Fig. S.2. The X-ray beam generated by an undulator source was monochromatized to a photon energy of 10.9 keV by a Ge(111) double crystal monochromator. Compound refractive lenses (CRL) at a distance of 66.9 m from the source with a focal length of 2.18 m provided vertical focusing. A 150 μm vertical aperture upstream of the CRL defined the vertical divergence to be 67 μrad FWHM. The horizontal beam size was defined by a 15 μm slit located 1.04 m upstream of the sample. The horizontal divergence, defined by the

645 μm FWHM undulator source size 69.4 m upstream of the sample, was 9.3 μrad FWHM. The sample was located downstream of the CRL focus, at a position giving a beam size of $15 \times 6 \mu\text{m}^2$ (h $\times$ v FWHM). Incident X-ray flux was monitored with an ion chamber placed upstream of the sample. The typical flux at the sample position was 2.3×10^{10} ph/s. The ion chamber and sample were in an air region of 31.5 cm length. Vacuum flight paths with Si_3N_4 (upstream, before sample), polycarbonate (downstream, after sample) and Kapton (downstream, before detector) windows were used to minimize scattering and absorption.

The sample heater consisted of insulated aluminum and copper blocks into which the sample capillary was inserted. Temperature was controlled using a Peltier element backed by water cooling. Temperature was determined with type K thermocouples with an uncertainty of ± 0.1 K from a combination of stability and accuracy. When changing temperature, it overshoot by about 10% of the temperature step. Near T_{sp} we used a step of 0.2 K, giving as overshoot of 0.02 K.

Scattered intensity was recorded 3.72 m downstream of the sample using a high-speed photon-counting area detector [9]. The detector has a rectangular matrix of 256×128 square pixels with a spacing of 75 μm . With a dynamic range from 0 to 3 photon counts per pixel per frame, the detector acquires sequences (“batches”) of frames at a frame rate of 50 kHz (20 μs per frame). Each batch consists of 2×10^5 frames (4 seconds of continuous acquisition), limited by the size of the memory on the detector controller. A gap of ~ 10 s is required between the batches to compress the sparse data in each frame and transfer the compressed frames from the memory of the detector controller to the network hard drive of the beamline. For each temperature, 150 to 400 batches were acquired for XPCS analysis. The transmitted primary beam was absorbed by a round tungsten beamstop placed in front of the detector.

Fig. S.3 shows typical examples of a single frame and of the average of one batch (2×10^5 frames). The averaged scattering pattern shows counts on the order of 10^{-3} photons per pixel per frame. The lack of azimuthal dependence shows that the intensity only depends on the modulus q of the wave vector. White areas in Fig. S.3 are masked pixels excluded from data analysis. These are pixels behind the beamstop and a few bad pixels. Batches of single frames, each with just a few single-photon events (Fig. S.3(a)), were used in XPCS analysis. The capability of this detector to perform such low-intensity, high-speed XPCS experiments has been demonstrated previously [9].

Other measurements

Dynamic light scattering (DLS) measurements were performed with Zetasizer Nano ZSP (Malvern Panalyti-

cal) equipped with laser of wavelength $\lambda_{\text{DLS}} = 632.3$ nm and temperature controller. The scattering angle was fixed at $2\theta_{\text{DLS}} = 175^\circ$. For DLS measurements, the sample was loaded into a square glass cuvette (Starna, 3-G-10, 10×10 mm).

The shear viscosity η_s was measured using a LOVIS 2000 M/ME Microviscometer (Anton Paar) in a T range from 298 K down to 294 K with step of 1 K. The refractive index of the sample for visible light at $\lambda = 632.3$ nm was measured with an Abbemat refractometer (Anton Paar Instruments) and found to be 1.430 ± 0.002 over the range of T studied.

FITTING METHODOLOGY

For all of the fits described below, we used a Levenberg-Marquardt nonlinear regression algorithm [10, 11]. This allows fitting to arbitrary functions $f(x_i; \mathbf{p})$, and finds the parameter values $\mathbf{p}$ that minimize a weighted chi-square $\chi^2 \equiv \sum_i ([y_i - f(x_i; \mathbf{p})]/\sigma_i)^2$, where the σ_i are the uncertainties in the data points to be fitted y_i . In addition to the parameter values that minimize χ^2 , the algorithm returns a covariance matrix of the parameters calculated from the dependence of χ^2 on $\mathbf{p}$ near the minimum. The one-standard-deviation uncertainties on the parameters that we report are the square root of the diagonal elements of the covariance matrix. This properly accounts for the larger uncertainty that can arise when there is correlation between the parameters.

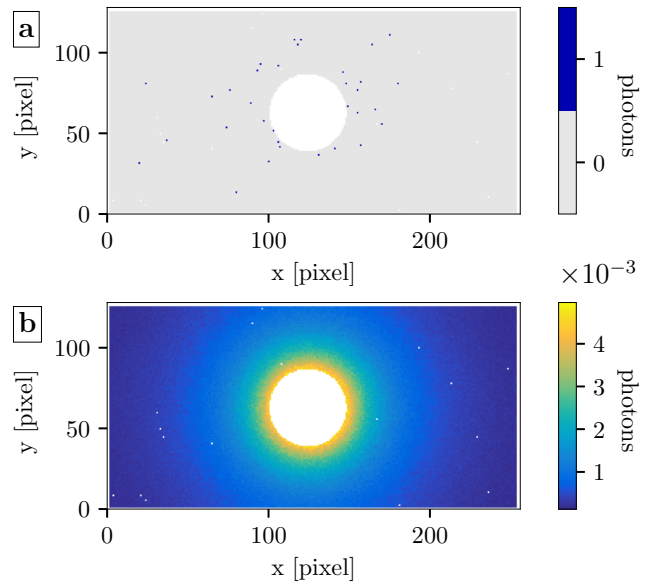


FIG. S.3. (a) Single scattering pattern and (b) averaged scattering pattern over 2×10^5 exposures for a temperature $T = 291.6$ K. White areas in (b) are masked pixels.

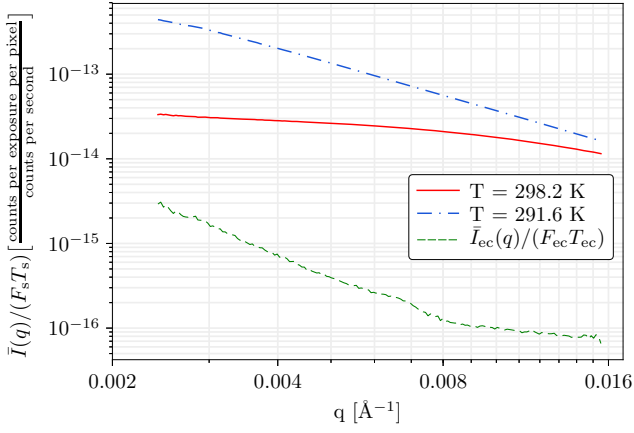


FIG. S.4. Normalized scattered intensity $\bar{I}_s(q)/F_s T_s$ as a function of wavenumber q for temperatures near and far from T_{sp} , together with the empty capillary background $\bar{I}_{ec}(q)/F_{ec} T_{ec}$.

TABLE S.1. Values of I_{q_0} and ξ obtained from 41-parameter fit to all 3600 time-averaged data points at 180 q and 20 T . The dimensional anomaly exponent obtained is $\eta = 0.064 \pm 0.001$.

Temperature, [K]	I_{q_0} [cm^{-1}]	ξ [$\text{\AA}$]
298.2	9.47 ± 0.01	85.89 ± 0.07
297.2	13.56 ± 0.01	103.05 ± 0.08
296.2	16.14 ± 0.01	113.49 ± 0.09
295.7	18.05 ± 0.01	120.68 ± 0.09
295.2	20.71 ± 0.01	129.43 ± 0.10
294.7	25.24 ± 0.02	143.72 ± 0.11
294.2	29.39 ± 0.02	156.13 ± 0.12
294.0	32.90 ± 0.03	165.45 ± 0.13
293.8	35.22 ± 0.03	171.23 ± 0.13
293.6	39.52 ± 0.03	182.48 ± 0.14
293.4	40.62 ± 0.03	185.16 ± 0.15
293.2	47.30 ± 0.04	200.69 ± 0.16
293.0	53.36 ± 0.05	213.76 ± 0.18
292.8	66.18 ± 0.07	239.38 ± 0.22
292.6	85.04 ± 0.11	273.34 ± 0.27
292.4	102.01 ± 0.14	301.05 ± 0.33
292.2	129.81 ± 0.21	341.18 ± 0.42
292.0	176.16 ± 0.36	401.23 ± 0.58
291.8	212.81 ± 0.50	443.53 ± 0.73
291.6	593.83 ± 3.39	766.55 ± 2.65

ANALYSIS OF TIME-AVERAGED DATA

Absolute scale of intensity profiles

To obtain the static critical behavior as a function of temperature, scattering patterns at each T were time averaged over the full range of batches recorded. The resulting 2D patterns were corrected for detector flat field and shadowing (determined by response to a uniformly scattering sample) and azimuthally averaged to obtain

intensity profiles $\bar{I}(q)$ as a function of wave vector modulus q at each T . Azimuthal averaging was done by partitioning the pixels into 180 groups at equivalent q , equally spaced in $\log(q)$, and averaging each partition. In the main text in Fig. 2(a) only selected temperatures and wavenumbers q are shown. The background scattering measured using an empty capillary was normalized and subtracted, and the intensity on an absolute scale $I(q)$ (scattering cross section per unit volume, cm^{-1}) was obtained using

$$I(q) = \left[\frac{\bar{I}_s(q)}{F_s T_s} - \frac{\bar{I}_{ec}(q)}{F_{ec} T_{ec}} \right] \frac{1}{t_{\text{exp}} f \Delta\Omega}, \quad (\text{S.1})$$

where $\bar{I}_s$ ($\bar{I}_{ec}$) is the azimuthally averaged scattering (in units of counts per exposure per pixel) from the sample (empty capillary), F_s (F_{ec}) is the flux (in units of counts per second) incident on sample (empty capillary), $T_s = 0.49$ ($T_{ec} = 0.83$) is the measured transmission of the sample (empty capillary), $t_{\text{exp}} = 20 \mu\text{s}$ is the exposure time, $f = 4 \text{ mm}$ is the sample thickness, and $\Delta\Omega = (75\mu\text{m}/3.72\text{m})^2$ is the solid angle subtended by a detector pixel. Fig. S.4 shows $\bar{I}_s(q)/F_s T_s$ for temperatures near and far from T_{sp} , together with the empty capillary background $\bar{I}_{ec}(q)/F_{ec} T_{ec}$ that was subtracted.

Static scaling fits

Static scaling laws give the expressions

$$I(q, T) = \frac{I_{q_0}(T)}{1 + x^{2-\eta}}, \quad (\text{S.2})$$

$$I_{q_0}(T) = I_0 [(T - T_{sp})/T_{sp}]^{-\gamma}, \quad (\text{S.3})$$

$$\xi(T) = \xi_0 [(T - T_{sp})/T_{sp}]^{-\nu}, \quad (\text{S.4})$$

where $x \equiv q\xi(T)$ is the reduced wavenumber and $\xi(T)$ is the correlation length, T_{sp} is the spinodal temperature, η is the dimensional anomaly exponent, and I_0 and ξ_0 , γ and ν are the amplitudes and critical exponents for I_{q_0} and ξ , respectively.

The $I(q)$ for all 20 temperatures were simultaneously fit with Eq. (S.2) with 41 parameters: I_{q_0} and ξ for each T and a single exponent η . We fit to $\log(I)$ with equal weighting, because we expect the primary contribution to noise in $I(q)$ is not photon counting statistics but is systematic noise proportional to the signal e.g. residual detector nonuniformity of response. We did a simultaneous fit of all temperatures, that allows the value of η that is best determined near T_{sp} to constrain the fit far from T_{sp} . If use this value of η and fit each T separately varying only I_{q_0} and ξ , we obtain the same values for $I_{q_0}(T)$ and $\xi(T)$ as in the simultaneous fit. Obtained fit parameters I_{q_0} and ξ with their uncertainties are presented in Table S.1.

TABLE S.2. Parameters obtained from time-averaged data using different fitting procedures (see text).

Fit type	η	T_{sp} [K]	ν	γ	ξ_0 [Å]	I_0 [cm ⁻¹]
I (41 parameters)	0.064 ± 0.001					
fit of I_{q_0}		291.26 ± 0.11		1.24 ± 0.06		0.10 ± 0.02
fit of ξ		291.31 ± 0.08	0.64 ± 0.03		8.3 ± 0.9	
II (5 parameters)	0.084 ± 0.004	291.22 ± 0.01	0.645 ± 0.002	$(2 - \eta)\nu$	8.3 ± 0.1	0.104 ± 0.002

The resulting I_{q_0} and ξ were then fit using Eq. (S.3) and Eq. (S.4), respectively, to obtain the amplitudes, exponents, and T_{sp} . We used the uncertainties of I_{q_0} and ξ in Table S.1 to give the weighting. The best-fit values for η , T_{sp} , ν , γ , ξ_0 , I_{q_0} are presented in Table S.2. The obtained best-fit values of T_{sp} from both fits, I_{q_0} and ξ , agree within their error. An average value of $T_{\text{sp}} = 291.3 \pm 0.1$ K was used in subsequent analysis.

In addition to the static data fit with 41 parameters, data for all 20 temperatures were fit using equations (S.2)-(S.4) with 5 parameters: η , T_{sp} , I_0 , ξ_0 , ν . The 5-parameter fit assumes that critical exponent for susceptibility follows the relation $\gamma = (2 - \eta)\nu$, while in the 41-parameter fit this relation was not forced. The obtained best-fit values for all parameters except η from both fits agree within their error (see Table S.2). Values from the 41-parameter fit are used in the main paper.

Nature of critical fluctuations

The agreement of the critical exponents we observe with those predicted for a binary fluid system indicates that the critical fluctuations are predominantly variations along a single direction in composition space. Based on the phase diagram in Fig. 1 of the main paper, we expect that the unstable composition coordinate is the concentration of micelle-like structures containing fixed fractions of aqueous and extractant components, shown by the pseudo-binary cut. Since the X-ray scattering intensity from composition fluctuations is proportional to the square of the difference in electron density of the end-members of the pseudo-binary, the question arises whether the scattering is dominated by the “heavy” ionic species in $\text{Ce}(\text{NO}_3)_3$. The composition of the aqueous-plus-extractant micelle end-member can be estimated from measurements of similar compositions in the literature [4], giving 0.6 moles of water, 0.375 moles of HNO_3 , and 0.02 moles of $\text{Ce}(\text{NO}_3)_3$ per mole of DMBTDMA. Its density can be estimated by assuming that the partial molar volumes are independent of composition. Comparing the electron densities of dodecane (0.4315 mole/cm³), micelles with an aqueous component just containing water and acid (0.6029 mole/cm³), and micelles also containing $\text{Ce}(\text{NO}_3)_3$ (0.6072 mole/cm³), there is only a 5% change in the scattering upon adding the $\text{Ce}(\text{NO}_3)_3$. So we expect that most of the scattering contrast comes

from the extractant, water, and acid, not from $\text{Ce}(\text{NO}_3)_3$. This estimate is in agreement with the small change in the mass density of the critical composition that we have measured (0.791 to 0.790 g/cm³) when reducing the $\text{Ce}(\text{NO}_3)_3$ concentration by a factor of two.

XPCS ANALYSIS

Extraction of $g^{(2)}$

The use of an area detector allows a “multi-speckle” analysis scheme where information from many pixels at equivalent q values can be averaged to improve the signal-to-noise ratio in the extracted $g^{(2)}$, allowing analysis of data with shorter exposure times and thus fewer photons per frame. The autocorrelation function $g^{(2)}$ is calculated using [9]

$$g^{(2)}(\Delta t) = \frac{\langle \langle I_p(t) I_p(t + \Delta t) \rangle_t \rangle_p}{\langle \langle I_p(t) \rangle_t \rangle_p \langle \langle I_p(t + \Delta t) \rangle_t \rangle_p} \quad (\text{S.5})$$

where $I_p(t)$ is intensity measured by a detector pixel p at time t , $\langle \rangle_t$ denotes the average over all time pairs within a batch and all batches, and $\langle \rangle_p$ is an average over all pixels at equivalent q . For these pixel averages we used the same 180 partitions defined for the static analysis. A multi-tau algorithm was used for the time averages [12, 13]. After calculation of $g^{(2)}(\Delta t)$ using Eq. (S.5) for the 180 partitions, we subsequently binned them by a factor of 10 to obtain results at 18 mean values of q which were equally spaced on a log scale. Fig. S.5 shows part of a scattering pattern with marked pixels in some of the 180 partitions used in Eq. (S.5) as well as the larger regions into which these were binned. Owing to low signal levels at higher q , only the 10 regions at lowest q were used to obtain $\tau(q)$.

Exponential fit of $g^{(2)}$ to obtain $\tau(q, T)$

To obtain the relaxation times τ , the $g^{(2)}(\Delta t)$ data at each q and T with sufficient signal were fit with a simple exponential function

$$g^{(2)}(\Delta t) = B + \beta \exp(-2\Delta t/\tau), \quad (\text{S.6})$$

TABLE S.3. Values of τ in μs and their uncertainties obtained from exponential fit of $g^{(2)}$ with Eq. (S.6) using fixed contrast $\beta = 0.093$ and free baseline B .

T [K]	q [$\text{\AA}^{-1}$]	0.0026	0.0028	0.0031	0.0035	0.0039	0.0043	0.0047	0.0053	0.0058	0.0065
296.2		223 $\pm$ 45	186 $\pm$ 28	187 $\pm$ 22	142 $\pm$ 20	125 $\pm$ 15	88 $\pm$ 15	104 $\pm$ 18	73 $\pm$ 10	45 $\pm$ 11	43 $\pm$ 9
295.7		362 $\pm$ 44	253 $\pm$ 20	190 $\pm$ 16	146 $\pm$ 11	118 $\pm$ 9	100 $\pm$ 11	84 $\pm$ 7	72 $\pm$ 8	60 $\pm$ 7	33 $\pm$ 8
295.2		331 $\pm$ 37	261 $\pm$ 18	247 $\pm$ 16	179 $\pm$ 13	127 $\pm$ 7	110 $\pm$ 11	84 $\pm$ 8	60 $\pm$ 8	54 $\pm$ 6	46 $\pm$ 8
294.7		358 $\pm$ 34	325 $\pm$ 20	242 $\pm$ 17	191 $\pm$ 12	152 $\pm$ 8	119 $\pm$ 7	93 $\pm$ 6	69 $\pm$ 7	62 $\pm$ 7	55 $\pm$ 6
294.2		349 $\pm$ 36	308 $\pm$ 21	270 $\pm$ 14	233 $\pm$ 11	182 $\pm$ 9	142 $\pm$ 8	107 $\pm$ 7	75 $\pm$ 8	56 $\pm$ 7	46 $\pm$ 7
294.0		525 $\pm$ 45	391 $\pm$ 24	257 $\pm$ 16	229 $\pm$ 16	179 $\pm$ 12	126 $\pm$ 10	99 $\pm$ 9	79 $\pm$ 9	82 $\pm$ 9	49 $\pm$ 8
293.8		395 $\pm$ 46	370 $\pm$ 18	330 $\pm$ 18	258 $\pm$ 16	180 $\pm$ 14	134 $\pm$ 13	103 $\pm$ 9	102 $\pm$ 9	62 $\pm$ 8	37 $\pm$ 7
293.6		556 $\pm$ 41	398 $\pm$ 24	354 $\pm$ 16	248 $\pm$ 15	228 $\pm$ 14	162 $\pm$ 12	133 $\pm$ 9	82 $\pm$ 7	67 $\pm$ 8	59 $\pm$ 8
293.4		468 $\pm$ 36	409 $\pm$ 22	357 $\pm$ 15	244 $\pm$ 11	183 $\pm$ 13	160 $\pm$ 11	136 $\pm$ 10	102 $\pm$ 10	63 $\pm$ 6	47 $\pm$ 8
293.2		580 $\pm$ 35	478 $\pm$ 21	386 $\pm$ 18	280 $\pm$ 13	225 $\pm$ 15	170 $\pm$ 11	134 $\pm$ 10	93 $\pm$ 8	78 $\pm$ 8	56 $\pm$ 8
293.0		555 $\pm$ 37	509 $\pm$ 23	383 $\pm$ 14	301 $\pm$ 14	234 $\pm$ 10	172 $\pm$ 9	126 $\pm$ 8	99 $\pm$ 8	86 $\pm$ 9	56 $\pm$ 6
292.8		650 $\pm$ 33	533 $\pm$ 17	434 $\pm$ 13	347 $\pm$ 13	267 $\pm$ 11	194 $\pm$ 9	138 $\pm$ 8	94 $\pm$ 7	85 $\pm$ 9	61 $\pm$ 5
292.6		730 $\pm$ 30	617 $\pm$ 14	491 $\pm$ 15	371 $\pm$ 13	294 $\pm$ 12	208 $\pm$ 8	160 $\pm$ 6	101 $\pm$ 6	91 $\pm$ 6	67 $\pm$ 7
292.4		745 $\pm$ 25	657 $\pm$ 17	497 $\pm$ 12	383 $\pm$ 10	281 $\pm$ 11	208 $\pm$ 8	164 $\pm$ 8	110 $\pm$ 7	74 $\pm$ 5	65 $\pm$ 6
292.2		1004 $\pm$ 33	802 $\pm$ 19	586 $\pm$ 13	446 $\pm$ 8	331 $\pm$ 10	231 $\pm$ 7	167 $\pm$ 6	127 $\pm$ 5	88 $\pm$ 6	59 $\pm$ 6
292.0		978 $\pm$ 30	781 $\pm$ 15	584 $\pm$ 11	440 $\pm$ 9	318 $\pm$ 8	239 $\pm$ 7	176 $\pm$ 8	124 $\pm$ 5	91 $\pm$ 5	65 $\pm$ 5
291.8		933 $\pm$ 24	783 $\pm$ 14	602 $\pm$ 10	432 $\pm$ 8	296 $\pm$ 7	223 $\pm$ 7	173 $\pm$ 6	124 $\pm$ 6	99 $\pm$ 6	60 $\pm$ 5
291.6		1043 $\pm$ 11	825 $\pm$ 9	624 $\pm$ 8	447 $\pm$ 8	320 $\pm$ 7	220 $\pm$ 6	164 $\pm$ 5	127 $\pm$ 6	87 $\pm$ 5	60 $\pm$ 6

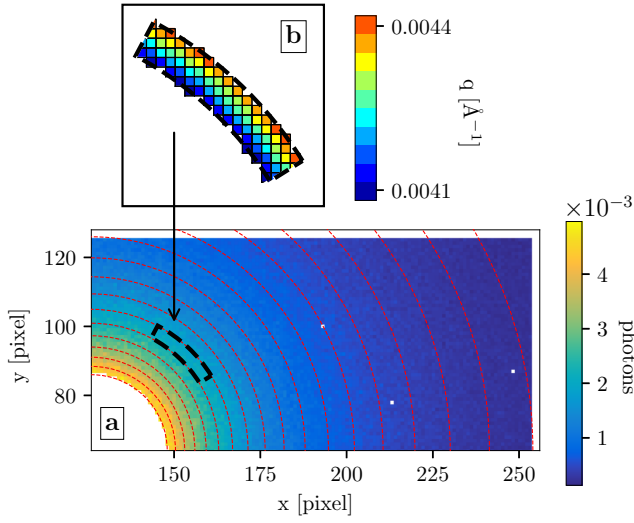


FIG. S.5. (a) A quadrant of a scattering pattern overlaid with 18 annular regions into which the $g^{(2)}$ results were binned. (b) Zoomed-in area of the 6th region with some of the 180 partitions used in the static analysis and the $\langle g \rangle_p$ averages in Eq. (S.5). Colors denote q values and are indicated in colorbar.

where β is a sample-independent contrast parameter related to the coherence properties of the X-ray beam and the scattering geometry and B is a baseline parameter close to unity.

Signal was sufficient to obtain τ for the 18 temperatures closest to T_{sp} and the 10 wavenumber regions out to $q = 0.0065 \text{ \AA}^{-1}$. For small-angle scattering with

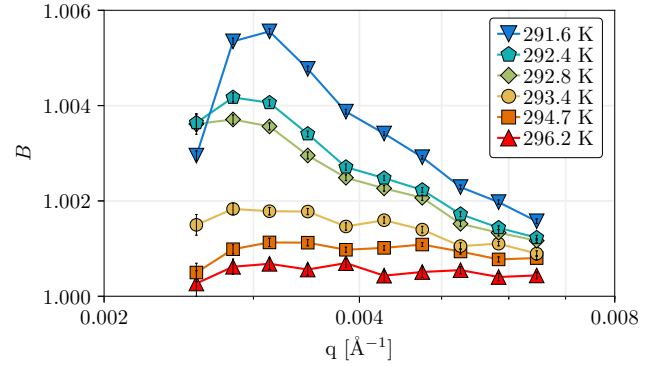


FIG. S.6. Fit values of the baseline B , as a function of wavenumber q for different temperatures.

monochromatic beam, the contrast β is independent of q [14]. It depends only on beamline parameters such as incident beam coherence and detector angular resolution. We used a value of $\beta = 0.093$ which is an average of values determined for the setup used. We checked the dependence of the results on using different contrast values $\beta = 0.086$ and $\beta = 0.100$. This amount of contrast difference shifts all τ values by $\pm 7\%$ from the values reported, approximately uniformly over all T and q . It thus affects the values of R obtained from the fits of D^* to Ω and Ω_{emp} , but has little effect on the comparison between the chi square of the two fits. Because the average intensity $I(q)$ can vary within each partition of pixels used in the $\langle g \rangle_p$ average in Eq. (S.5), the baseline parameter B can deviate from unity and be a function of T and q , reflecting the static variance in the average inten-

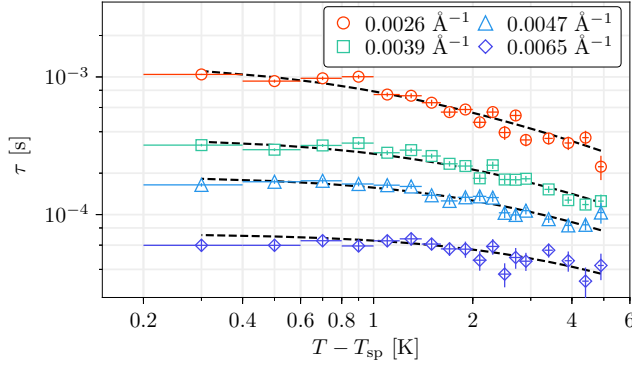


FIG. S.7. Extracted τ as a function of T at selected q . Dashed lines are fits with Eq. (5) in the main text.

sity within each partition. Thus τ and B were varied in the fits, with $g^{(2)}$ values weighted according to their uncertainties obtained from the multi-speckle analysis. Best-fit values of τ and B are shown in Table S.3 and in Fig. S.6, respectively. Figure S.7 shows τ as a function of T for selected q .

The relaxation times τ at each T were fit with a power law $\tau \propto q^{-z_{\text{eff}}}$ to obtain values of z_{eff} , as shown in Fig. 3 of the main paper. For these fits the values of τ were weighted according to their uncertainties reported in Table S.3.

Viscosity and “background” diffusivity

The reduced diffusivity D^* is obtained from the measured diffusivity $D \equiv \tau^{-1}q^{-2}$ using

$$D^*(q, T) = \frac{D(q, T) - \bar{D}(q, T)}{D_0(T)}. \quad (\text{S.7})$$

We use the standard expressions for D_0 and $\bar{D}$ [15]

$$D_0(T) = \frac{k_B T}{6\pi\eta_s(T)\xi(T)}, \quad (\text{S.8})$$

$$\bar{D}(q, T) = \frac{k_B T}{16\bar{\eta}_s(T)\xi(T)} \left(\frac{1 + [q\xi(T)]^2}{q_C\xi(T)} \right). \quad (\text{S.9})$$

To evaluate $D_0(T)$, we used measured values of the shear viscosity η_s , interpolated or extrapolated in T using a fit function described below. To evaluate the “background” (non-critical) diffusivity $\bar{D}(q, T)$, we also need values of the “background” viscosity $\bar{\eta}_s(T)$ and the cutoff wavenumber q_C . The relation between $\eta_s(T)$ and $\bar{\eta}_s(T)$ is [15]

$$\eta_s(T) = \bar{\eta}_s(T) \exp[z_\eta H(T)], \quad (\text{S.10})$$

where the critical exponent $z_\eta = 0.063$ is the same as that introduced in the definition of Ω in the main paper, and

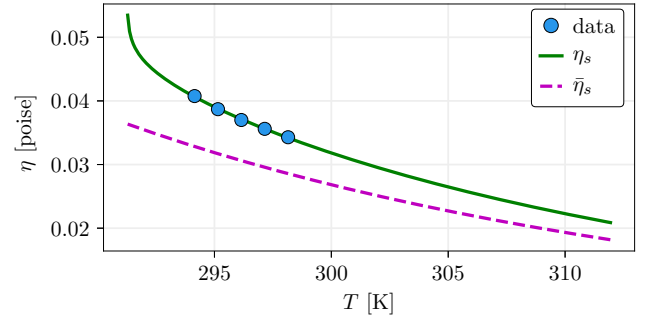


FIG. S.8. Viscosity as a function of T . Markers are measured values; solid curve is fit with Eq. (S.10); dashed curve is the background viscosity $\bar{\eta}_s$.

the function $H(T)$, specified in [15], depends on $\xi(T)$, q_C , and a second cutoff wavenumber q_D which we take to be $q_D = 1.7q_C$ [15]. To obtain $\bar{\eta}_s(T)$, we fit the measured viscosity data to Eq. (S.10), using an Arrhenius form for $\bar{\eta}_s(T)$,

$$\bar{\eta}_s = \eta_0 \exp(E_\eta/k_B T), \quad (\text{S.11})$$

where η_0 and E_η were the two parameters varied. The fit is shown in Fig. S.8, for a cutoff wavenumber of $q_C = 0.56 \times 10^8 \text{ cm}^{-1}$. The best fit values of the parameters were $\eta_0 = 1.0 \times 10^{-6} \text{ Poise}$ and $E_\eta = 0.26 \text{ eV}$. The curves shown for $\eta_s(T)$ and $\bar{\eta}_s(T)$ were used in the expressions for $D_0(T)$ and $\bar{D}(q, T)$, respectively. We use values from the power-law fit result $\xi(T) = \xi_0[(T - T_{\text{sp}})/T_{\text{sp}}]^{-\nu}$, when evaluating $D_0(T)$, $\bar{D}(q, T)$, and $H(T)$.

The value of the cutoff wavenumber q_C influences the size of the background term $\bar{D}(q, T)$, with a smaller q_C giving a larger $\bar{D}$. Several previous DLS studies of complex fluids containing micelles [16–19], microemulsions [20–23], or macromolecules [24] have found the need to use small values of q_C , and thus large $\bar{D}(q, T)$, to get agreement with dynamic scaling [25, 26]. We tested the effect of using different values of q_C on the collapse of our D^* values onto a single master curve. Table S.4 summarizes the effect of q_C on the χ^2 of the fit of $D^*(x)$ to $\Omega(x)$, which characterizes the deviation of the $D^*(x)$ from a single master curve. For our system, we find that the best dynamic scaling occurs using large values of q_C that give negligible $\bar{D}$. We have therefore adopted the relatively large value of $q_C = 0.56 \times 10^8 \text{ cm}^{-1}$ that was used in [15]. The background term is a small contribution, with $\bar{D}/D$ ranging from 0.8% to 2.7% depending on T and q (see Fig. S.9).

Fitting of Ω and Ω_{emp} to D^*

The $\log(D^*)$ values for all 180 pairs of T and q values were fit to $\log(\Omega(x))$ and $\log(\Omega_{\text{emp}}(x))$ from Eqs. (6) and

TABLE S.4. Comparison of dynamic scaling fits of Ω to D^* using different values of q_C .

$q_C [\text{cm}^{-1}]$	R	χ_r^2	$(\bar{D}/D)_{\min}$	$(\bar{D}/D)_{\max}$
1.00×10^8	0.91 ± 0.06	4.1	0.5×10^{-2}	1.7×10^{-2}
0.56×10^8	0.90 ± 0.06	4.2	0.8×10^{-2}	2.8×10^{-2}
0.20×10^8	0.88 ± 0.06	4.4	2.2×10^{-2}	7.3×10^{-2}

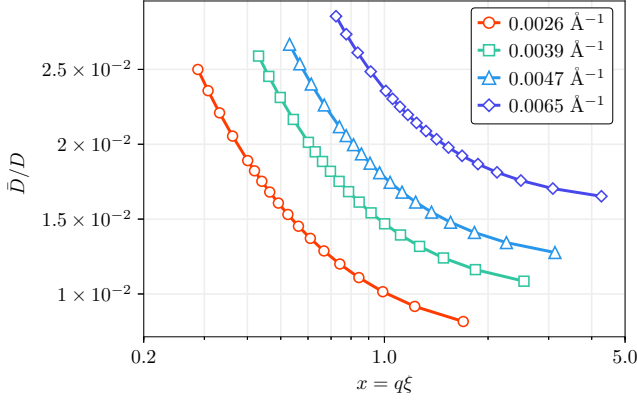


FIG. S.9. Relative background contribution $\bar{D}/D$ as a function of the normalized wavenumber x for different q , calculated for $q_C = 0.56 \times 10^8 \text{ cm}^{-1}$.

(9) in the main paper, respectively. The $x \equiv q\xi$ values for each point were calculated using $\xi(T)$ from Table S.4, and each point was weighted using the relative uncertainty in τ , ignoring the smaller relative uncertainty in D_0 . The fits are shown in the main paper Fig. 4, and the residuals of the fits are shown as D^*/Ω and D^*/Ω_{emp} in Fig. S.10. The values of D^*/Ω data deviate positively from unity at both low and high q , reflecting the less abrupt change in slope of $\Omega(x)$ relative to the data. The fit to Ω_{emp} is significantly better than that to Ω , with a reduced chi square of $\chi_r^2 = 2.5$ compared with $\chi_r^2 = 4.2$, respectively.

Comparison of XPCS and DLS at low q

We compared the fluctuation times observed by XPCS at low q to those given by dynamic light scattering (DLS). The wavenumber $q_{\text{DLS}} = 0.0028 \text{ \AA}^{-1}$ was obtained from

$$q_{\text{DLS}} = \frac{4\pi n}{\lambda_{\text{DLS}}} \sin(\theta_{\text{DLS}}). \quad (\text{S.12})$$

where $n = 1.430$ was the measured refractive index.

As in XPCS, the $g^{(2)}$ obtained with DLS were fit with a simple exponential decay in order to extract the relaxation time τ_{DLS} ,

$$g^{(2)}(\Delta t) = 1 + \beta_{\text{DLS}} \exp(-2\Delta t/\tau_{\text{DLS}}). \quad (\text{S.13})$$

The best fit value of the contrast parameter β_{DLS} was found to be 0.90 ± 0.02 . A comparison of the $T - T_{\text{sp}}$ de-

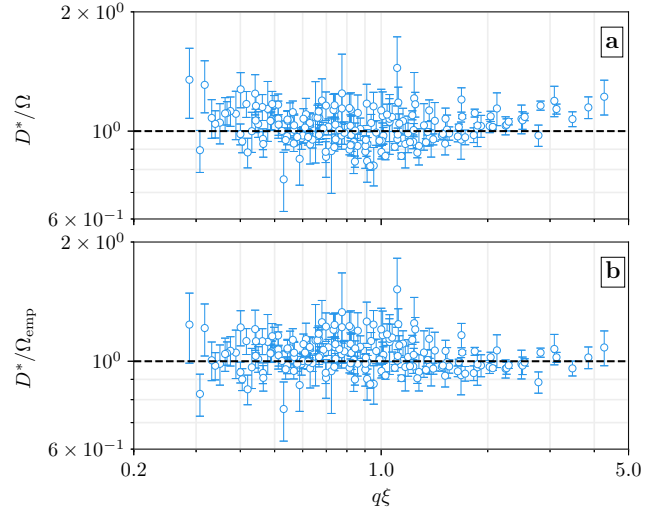


FIG. S.10. Residuals of fits to D^* , shown as ratio of D^* to (a) Ω and (b) Ω_{emp} , as a function of reduced wavenumber $q\xi$. Uncertainties shown are calculated from uncertainties in τ given in Table S.3.

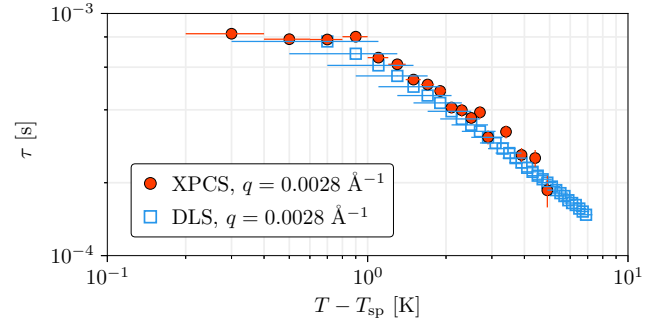


FIG. S.11. Comparison of relaxation times τ obtained by XPCS and by DLS as a function of temperature at $q = 0.0028 \text{ \AA}^{-1}$.

pendencies of the relaxation times obtained from XPCS and DLS at the same $q = 0.0028 \text{ \AA}^{-1}$ is shown in Fig. S.11. For T_{sp} we used the value of 291.3 K obtained from the time-averaged X-ray data. Measurement of dynamics close to T_{sp} was complicated in the DLS experiment because the temperature stability was only 0.4 K and because of non-uniform cooling of the sample due to the bigger size of the sample cell. Splitting of the organic phase was observed at a higher T in the DLS experiment than in the XPCS experiment. Nevertheless, the values of the relaxation time at $q = 0.0028 \text{ \AA}^{-1}$ obtained with DLS show agreement within 10% with XPCS data across the full T range.

X-ray interaction with sample behavior

One of the potential issues with XPCS is the possibility of undesirable interaction of the ionizing X-rays with the sample, potentially changing its chemistry or dynamics. We purposely placed the sample downstream of the focus of the CRL optics to reduce the X-ray flux density on the sample. If we use the $15 \times 6 \mu\text{m}^3$ beam size to calculate the dose from 2.3×10^{10} ph/s at 10.9 keV with a mass absorption coefficient $\mu_\rho = 1.8 \text{ cm}^2/\text{g}$, we obtain 1.6 Gy, 0.3 MGy, and 66 MGy per 20 μs exposure time, per 4 s batch, and per 200-batch typical dataset at each T , respectively. These calculated doses are large, since doses of less than 1 MGy have been found to affect XPCS measurements in other soft matter systems [27, 28].

We expect this LLE system to be relatively robust because of both its high ionic conductivity and the long diffusion length of any ionized species. For a diffusion radius of $\sqrt{Dt}$, the effective dosed area increases linearly with time t . Using a typical molecular diffusivity of $D = 10^{-5} \text{ cm}^2/\text{s}$ for a liquid, the area of ion diffusion πDt exceeds our beam area after a time of only 30 ms. The dose in that time is only 2.4 kGy, and it does not increase further with time since the effective area continues to increase linearly.

To quantify any beam interaction, we analyzed the changes in intensity during each batch in different regions of q . We observed relatively small changes in intensity with time at each T - typically less than 2%, with a maximum of 5%. Since these small changes did not affect the results, we did not employ further methods to minimize beam interaction (such as translating the sample during acquisition [28] or low dose x-ray speckle visibility spectroscopy [27]).

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